2024 AMC 10B Problem 7 - Solution

Review the full statement and step-by-step solution for 2024 AMC 10B Problem 7. Great practice for AMC 10, AMC 12, AIME, and other math contests

2024 AMC 10B Problem 7

Problem:
What is the remainder when $7^{2024}+7^{2025}+7^{2026}$ is divided by 19 ?

Answer Choices:

A. 0

B. 1

C. 7

D. 11

E. 18

Solution:

The quantity in question is seen to be a multiple of 19 as follows:

$$7^{2024}+7^{2025}+7^{2026}=7^{2024}\left(1+7+7^2\right)=7^{2024}\cdot 57=7^{2024}\cdot 3\cdot 19$$

Therefore the remainder when it is divided by 19 is $(\mathrm{A})\boxed{0}$.

$$\textbf{OR}$$

Working modulo $19, 7^0 = 1, 7^1 = 7 \cdot 1 = 7, 7^2 = 7 \cdot 7 = 49 \equiv 11$, and $7^3 \equiv 7 \cdot 11 = 77 \equiv 1$. Therefore the remainders when dividing successive nonegative powers of 7 by 19 repeat with period 3 . The sum of any three consecutive remainders is therefore $1 + 7 + 11 = 19$, so the remainder when the given sum is divided by 19 is $(\mathrm{A})\boxed{0}$.